

5 (a) Explain what is meant by a progressive transverse wave.

progressive
.....
transverse
..... [2]

(b) A 30 W light bulb, connected to a power source, emits light radially in all directions.

(i) Determine the intensity of the light of the bulb at a point 3.0 m away.

intensity = W m^{-2} [2]

(ii) Determine the ratio

$$\frac{\text{amplitude of light at a point 2.0 m away}}{\text{amplitude of light at a point 3.0 m away}}$$

ratio = [2]

(iii) A similar bulb is used in a torchlight, where there is a concave mirror placed around the bulb.

Explain how the intensity of the light from the torchlight at 3.0 m away differs from your answer in (b)(i).

.....
.....
..... [2]

6 Fig. 6.1 shows three equally spaced equipotential lines between two charged parallel